

An Introduction *to* Scaffolding *with* Yeoman

Scaffolding in the computing context refers to two techniques – one is for the generation of code related to database access in some model-view-controller frameworks, and the second is a project-generation technique supported by other tools. Yeoman is an open source scaffolding tool for Web applications that helps developers to get started on a project.



Scaffolding, also called staging, is a temporary structure used to support people working in high, inaccessible places, particularly on construction sites. Scaffolding, when used in computing, refers to one of the following two techniques:

- A code generation technique related to database access in some model-view-controller frameworks; or
- A project generation technique supported by various tools.

Code generation

In the technique supported by some model-view-controller frameworks, the programmer can specify how the application database may be used. The compiler or framework uses this specification, together with pre-defined code templates, to generate the final code that the application can use to create, read, update and delete database entries, effectively treating the templates as a *scaffold* on which to build a more powerful application.

Scaffolding can occur at two different phases of the program's life cycle—at design time and runtime. Design

time scaffolding produces files of code that can later be modified by the programmer to customise the way the application database is used. Runtime scaffolding produces code on-the-fly. It allows changes to the design of the templates to be immediately reflected throughout the application.

Scaffolding techniques based on the application database typically involve server-side frameworks. Server-side Web frameworks commonly perform operations directly against database entries, and code generation for these operations may be considered server-side scaffolding. Client-side development often uses frameworks that perform data transport operations instead of directly accessing the database. The focus of client-side scaffolding is thus more on generating a starter template for the application, as a whole, rather than generating code to access a database.

Project generation

Complicated software projects often share certain conventions on project structure and requirements.